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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 03**

Total No. of Questions : 09

B.Tech. (CE) (Sem.–6)

DESIGN OF CONCRETE STRUCTURES

Subject Code : PECE-603A-18

M.Code : 79400

Date of Examination: 19-12-2022

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

1. Write briefly :

a) Write a short note on Substitute Frame Method.

b) Write down two advantages of Flat slab.

c) What are different types of liquid retaining structures?

d) Write a note on Equivalent Frame method.

e) What are curved beams?

f) Write a note on vibration Characteristics of Foundation.

g) Differentiate between cantilever and counterfort retaining wall.

h) Explain basic action of flat slab.

i) Name the different methods for the analysis of frames under lateral loads.

j) Write down design steps of raft foundation.

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SECTION-B

2. Design the stem of a cantilever retaining wall for the following requirements:

   Height of wall above ground level -5 m

   Superimposed load due to road traffic -18 kN/m^2

   Unit weight of fill -18 kN/m^2

   Angle of internal friction for fill material -28°

   Allowable bearing pressure on ground: 150 kN/m^2

   Coefficient of friction between concrete and ground: 0.4

   Height of parapet wall on top of stem:1 m

   Use M-20 concrete and Fe-415 grade steel.

3. A flat slab is supported on 600 mm dia circular columns spaced 8m × 6m apart in both directions. The column head has a diameter of 120 cm. The live load on the flat slab is 10kN/m^2 . Determine the moments in the flat slab along its 8m span and check the flat slab in shear at an edge column. Also tell the advantages and disadvantages of flat slab.

4. Write a detailed note on the types of Foundation.

5. Design a circular tank to the following requirements:

   (a) Diameter of tank = 6m, (b) Depth of water = 3.75m, (c) The tank rests on ground, (d) The walls and the base slab are not monolithic. Use M-20 concrete and Fe-415 steel.

6. What are the recommendations of IS 456 regarding redistribution of moment in the design of statically indeterminate structures employing working stress and limit state method?

SECTION-C

7. Explain the Portal and Cantilever method for analysing a building frame subjected to horizontal forces.

8. Design a Counterfort retaining wall with the following data:

   a) Height of embankment above ground is 10m.

   b) Depth of foundation is 1.5 m and net bearing capacity of soil is 150 kN/m^2.

   c) Weight of earth is 17 kN/m^3 and angle of repose is 30°.

   Check stability of the wall. Design and sketch the reinforcement details in the

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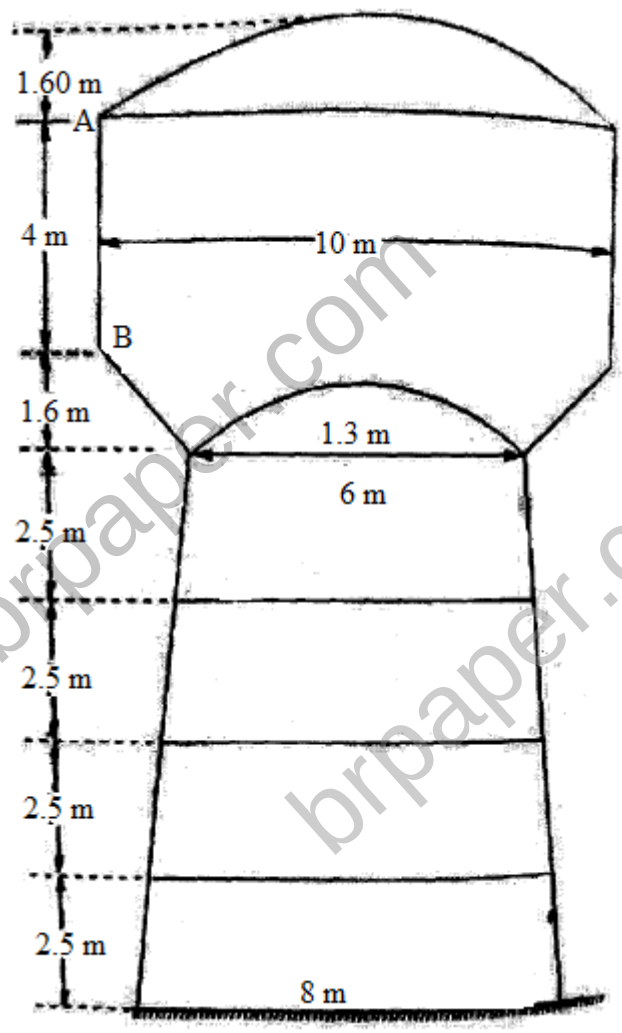
(i) Toe (ii) Heel (iii) Vertical slab (iv) Counterforts.

Design the ties to connect counterforts with vertical slab and heel slab.

9. Design an intze tank to the dimensions shown in figure. Use M-20concrete and Fe 250 steel.



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